

Unless otherwise specified, the groups are finite and the representations are finite dimensional in this text.

Week 1

1. Basic definition 1

- (a) What is a linear representation of a group? What is a G -module? What is the relationship between them?
- (b) What is a matrix representation of a group?
- (c) How to construct a matrix representation from a linear representation of G ? What is the relationship between different matrix representations induced from the same linear representation of G ?
- (d) If G is a finite abelian group, given a linear representation of G , how to choose a basis to let the corresponding matrix representation as simple as possible? (hint: simultaneous diagonalization)
- (e) Calculate permutation matrices for all $\sigma \in S_3$ (page 3)

2. Basic definition 2

- (a) What is an algebra over a field? (You can turn to Wikipedia if you have not met this concept yet.)
- (b) Describe the $\mathbb{C}$ -algebra structure of $GL(V)$ for a $\mathbb{C}$ -linear space V .
- (c) What is a linear representation of an algebra?
- (d) Let $G = \{g_1, g_2, \dots, g_n\}$ be a finite group. Let $\mathbb{C}[G]$ be the n -dim linear space spanned by $\{g_1, g_2, \dots, g_n\}$. $\mathbb{C}[G]$ has a natural $\mathbb{C}$ -algebra structure induced by the group structure by G :

$$\sum_{i=1}^n a_i g_i \cdot \sum_{j=1}^n b_j g_j = \sum_{k=1}^n \left(\sum_{g_i g_j = g_k} a_i b_j \right) g_k \quad \forall a_i, b_i \in \mathbb{C}$$

Check that $\mathbb{C}[G]$ is an algebra briefly. What is the relationship between G -mods and $\mathbb{C}[G]$ -mods?

- (e) $\mathbb{C}[G]$ also has a natural (left) G -module structure induced by the natural embedding from G to $\mathbb{C}[G]$. Describe this structure. If $G = S_3$, calculate the corresponding matrix representation by choosing a suitable basis.

3. One-dim representation

- (a) Explain what is a one-dim representation of G on $\mathbb{C}$ from the definition of linear representations. Please give a nontrivial example of it.
- (b) If G is a finite abelian group, determine all of the one-dim representations of G on $\mathbb{C}$. (You can admit the structure thm of finite abelian group after explaining what it is.) How many are they?

- (c) If G is not necessarily abelian, how to transform the question?
- (d) What may happen when we considering the one-dim representation of G on arbitrary fields? Can you give some examples?

Week 2

1. Reducibility, Maschke's theorem

- (a) Let V is a G -mod. Explain what is a G -submod of V and what is trivial G -submod.
- (b) State what is Maschke's theorem. Make comparison with linear space and show how the same claim works in linear space.
- (c) **Briefly** describe the first proof of the theorem. You need not to explain every details in the proof.
- (d) **Explicitly** repeat the second proof of the theorem. Explain what is a G -invariant inner product and how to construct a G -invariant inner product from an arbitrary inner product.
- (e) The second proof is much simpler than the first proof. What are the advantages of the first proof?
- (f) Let $G = S_3$, $V = \mathbb{C}m_1 \oplus \mathbb{C}m_2 \oplus \mathbb{C}m_3$, G act on V naturally. Suppose $U = \mathbb{C}(m_1 + m_2 + m_3)$. Find a G -submod W of V such that $V = U \oplus W$.

2. G -homomorphism and more

- (a) What is a indecomposable G -mod? Give a example of G -mod that is indecomposable but not irreducible. (Page 8)
- (b) Explain what is a irreducible G -mod and what is a direct sum of G -mod. Show that every G -mod can be decomposed into direct sum of irreducible G -mod. (hint: Use Maschke's theorem)
- (c) Rewrite the above proposition (including Maschke's theorem) in matrix terms.
- (d) Let V and W be G -mods. What is a G -homomorphism $\theta : V \rightarrow W$? What is a G -isomorphism? Let $\text{Hom}_G(V, W)$ be the set of G -hom from V to W . Then $\text{Hom}_G(V, W)$ has a natural $\mathbb{C}$ -linear space structure. Define the left action of G on $\text{Hom}_G(V, W)$ by

$$(g\theta)(v) = \theta(g^{-1}v), \quad \forall g \in G, \theta \in \text{Hom}_G(V, W), v \in V$$

Check that $\text{Hom}_G(V, W)$ is a G -mod.

- (e) Let V be a G -mod, X be the corresponding matrix representation. Define the invariant G -submod of V as

$$V^G = \{v \in V \mid gv = v, \forall g \in G\}$$

Prove that

$$\dim V^G = \frac{1}{|G|} \sum_{h \in G} \text{Tr}(X(h))$$

(hint: V^G is a G -submod of V and

$$p(v) = \frac{1}{|G|} \sum_{h \in G} hv, \quad \forall v \in V$$

is a projection to V^G .)

3. FIT and Schur's Lemma

- (a) Let V and W be G -mods and let $\theta : V \rightarrow W$ be a G -hom. Then prove that

$$\ker \theta := \{v \in V \mid \theta(v) = 0\}$$

is a G -submod of V and

$$\text{im } \theta := \{\theta(v) \mid v \in V\}$$

is a G -submod of W .

- (b) Prove that we have a canonical isomorphism of G -mod:

$$\bar{\theta} : V/\ker \theta \rightarrow \text{im } \theta$$

such that

$$\bar{\theta}(v + \ker \theta) = \theta(v), \quad \forall v \in V.$$

- (c) Explain and proof the three versions of Schur's lemma and the cor. one by one. (Page 10,11)
- (d) Use Schur's lemma to prove that every irreducible representation of an abelian group G is one-dimensional. (hint: version 3)
- (e) If G is a finite group, check that (G -module, G -homomorphism) forms an abelian category. (If you didn't know anything about category, feel free to ignore this.)

Week 3

1. Orthogonality relation

- (a) Proof the lemma in Page 11.
- (b) Explain and proof the orthogonality relation by direct calculation. (This is demanding, you should do the calculation carefully.)

2. Coordinate space of a representation and more

- (a) Explain what is V_G and what is the coordinate space for a representation S . Proof that equivalent $\mathbb{C}$ -matrix representations have the same coordinate space.

- (b) (Weyl's unitary trick) For an arbitrary matrix representation of a group G , show that there exists an invertible matrix T such that $TR(g)T^{-1}$ is a unitary matrix, $\forall g \in G$.
- (c) Suppose that all of $R^{(i)}$ s are unitary, proof (O4).
- (d) Under the same assumption as above, construct an inner product on $\mathbb{C}\mathcal{A}$ such that $R_{i,j}^{(k)}$ becomes an orthogonal basis of $\mathbb{C}\mathcal{A}$ under the inner product. Deduce that the number of pairwise inequivalent irreducible representations is less than $|G|$. Here $\mathcal{A}$ is define by

$$\mathcal{A} := \left\{ R_{pm}^{(k)} \mid k \in \{1, \dots, s\}, p, m \in \{1, \dots, d_k\} \right\}$$

and $\mathbb{C}\mathcal{A}$ is the linear space spanned by $\mathcal{A}$.

3. Regular representation and irreducible representation and more

- (a) Prove that the coordinate space of the regular representation R of a finite group G is equal to V_G .
- (b) Recall that every irreducible representation appears in the decomposition of the regular representation. Deduce that $\mathbb{C}\mathcal{A} = V_G$ and $\sum_{k=1}^s d_k^2 = |G|$.
- (c) Follow the exercise in page 16 and 17, construct an natural isomorphism between V_G and $\mathbb{C}[G]$.

Week 4

1. Class functions and characters

- (a) What is a class function? What is a character of a matrix representation? Check that a character is a class function and equivalent representations have the same character.
- (b) The set of all class functions in V_G forms a vector subspace of V_G . Prove that the set of irreducible characters of a finite group is a basis of the subspace of all class functions in V_G . Deduce that for a finite group G , the number of PIIRs of G is equal to the number of conjugacy classes in G .

2. Express the regular representation in term of irreducible representations

- (a) Prove that for a character χ , $\chi(g^{-1}) = \overline{\chi(g)}$.
- (b) View V_G as a inner product space, check that $\{\chi^{(1)}, \dots, \chi^{(s)}\}$ is a basis of the set of all class functions. In particular, we can use inner product to determine whether two irreducible representation are equivalent or not.
- (c) Proof that each irreducible representation (resp. character) occurs as a constitution of the regular representation (resp. character) with multiplicity equal to its degree.

3. Character table and more

- (a) Explain what is a character table.
- (b) Proof (CR) and (CR2).
- (c) For $G = S_3$, calculate its character table. (hint: trivial map and sign map to $\mathbb{C}$ are two irreducible representations.)

Let G_1 and G_2 be two finite group. $G := G_1 \times G_2$. Given representation $\rho : G_1 \rightarrow GL_m(\mathbb{C})$ and $\varphi : G_2 \rightarrow GL_n(\mathbb{C})$. Let $V = M_{mn}(\mathbb{C})$. Define $\tau : G \rightarrow GL(V)$ by $\tau(g_1, g_2)(A) = \rho(g_1)A\varphi(g_2)^T$, $\forall A \in V$.
- (d) Check that τ is a representation of G .
- (e) Check that $\chi_\tau(g_1, g_2) = \chi_\rho(g_1)\chi_\varphi(g_2)$
- (f) Prove that when ρ and φ are irreducible, τ is also irreducible. (hint: calculate (χ_τ, χ_τ))
- (g) Proof that all of the irreducible representation of G is of this type. (hint: how many irreducible representation are there?)